Autonomous Competitive Intelligence Agent

Hackathon Proposal – Loyalty Program Competitive Intelligence System

Overview

This proposal presents an Autonomous Multi-Agent Competitive Intelligence System designed to analyze and compare loyalty programs using live web retrieval, AI-based extraction, sentiment analysis, and source-attributed reporting.

The system transforms fragmented loyalty program data into actionable business intelligence through autonomous agents, retrieval-augmented generation (RAG), and verification pipelines.

Problem Understanding

The major challenge is not summarization, but collecting reliable and continuously updated information from multiple fragmented sources such as official websites, blogs, terms and conditions pages, user reviews, Reddit discussions, and press releases.

The solution must:
• Retrieve live information from the web
• Prevent hallucinated AI outputs
• Provide source-attributed claims
• Normalize different loyalty programs into comparable formats
• Generate meaningful business intelligence

Proposed Solution

We propose “IntelLoyal AI”, an autonomous multi-agent platform that:
• Crawls live loyalty program data
• Extracts rewards, tiers, and benefits
• Verifies claims using multiple sources
• Performs side-by-side comparisons
• Generates structured intelligence reports
• Produces insights and recommendations

System Architecture

1. User enters loyalty program names
2. Orchestrator AI distributes tasks to specialized agents
3. Web Retrieval Agent gathers live data
4. Sentiment Agent analyzes customer discussions
5. Benefit Extraction Agent structures reward information
6. Verification Agent validates claims and sources
7. Comparison Agent generates insights and SWOT analysis
8. Report Generator produces downloadable intelligence reports

---

Multi-Agent Design

Agent 1 – Web Retrieval Agent
• Uses Tavily, SerpAPI, Playwright, BeautifulSoup
• Retrieves official and public web information

Agent 2 – Sentiment Analysis Agent
• Analyzes Reddit, travel forums, and reviews
• Detects complaints and customer satisfaction trends

Agent 3 – Benefit Extraction Agent
• Extracts cashback rates, tiers, travel benefits, expiry rules

Agent 4 – Verification Agent
• Cross-checks claims using multiple sources
• Assigns confidence scores and citations

Agent 5 – Comparison Agent
• Creates side-by-side comparisons and SWOT analysis

Agent 6 – Report Generation Agent
• Produces executive summaries, dashboards, and PDF reports

Innovative Features

1. Trust Score System
Each claim receives:
• Confidence score
• Freshness timestamp
• Number of supporting sources

2. Hidden Rule Detector
Detects restrictive clauses such as:
• Expiry conditions
• Redemption caps
• Blackout periods

3. Consumer Persona Matching
Recommends best loyalty programs for:

• Frequent travelers
• Students
• Families
• Business users

4. Trend Intelligence
Detects:
• Reward devaluation
• Partnership changes
• Competitive strategy shifts

Technology Stack

Frontend:
• React
• Next.js
• Tailwind CSS

Backend:
• FastAPI / Node.js

AI Layer:
• OpenAI APIs
• LangChain
• CrewAI / AutoGen

Retrieval:
• Tavily
• SerpAPI
• Playwright
• BeautifulSoup

Databases:
• PostgreSQL
• Pinecone / ChromaDB / FAISS

RAG Pipeline

The system uses Retrieval-Augmented Generation (RAG):

1. Crawl live web data
2. Chunk documents
3. Generate embeddings
4. Store embeddings in vector database
5. Retrieve relevant context
6. Generate grounded responses with citations

This architecture reduces hallucinations and improves factual reliability.

Expected Output

The platform generates:
• Executive summaries
• Reward structure comparisons
• Sentiment insights
• SWOT analysis
• Side-by-side feature comparison tables
• Source-attributed intelligence reports

Feasibility and Planning

Week 1:
• Web crawling
• Basic summarization
• Citation system

Week 2:
• Comparison engine
• Vector database integration
• Sentiment analysis

Week 3:
• Multi-agent orchestration
• Dashboard UI
• Trust score system
• Hidden-rule detector

Stretch Goals:
• Real-time monitoring
• Multilingual support
• Enterprise dashboards

Risk Analysis

Risk 1 – Dynamic websites blocking scraping
Solution:
• Playwright browser automation
• API fallbacks

Risk 2 – Hallucinated claims
Solution:
• Mandatory citation verification
• Confidence scoring

Risk 3 – API rate limits
Solution:
• Caching
• Async retrieval pipelines

Team Composition

Member 1 – AI/LLM Engineer
• RAG pipeline
• Agent orchestration

Member 2 – Backend Engineer
• APIs
• Database integration

Member 3 – Frontend Engineer
• Dashboard UI
• Visualization

Member 4 – Data Engineer
• Scraping pipelines
• Data normalization

Demo Flow

Demo Example:
“Compare Marriott Bonvoy vs Hilton Honors”

System Workflow:
1. Retrieves live data
2. Extracts benefits
3. Verifies claims
4. Analyzes sentiment
5. Generates comparison report
6. Displays trust scores and recommendations

Conclusion

Our system transforms fragmented loyalty program information into actionable, trustworthy competitive intelligence using autonomous multi-agent AI, retrieval-augmented generation, and verifiable source attribution.

The platform is scalable, enterprise-ready, and designed for continuous competitive monitoring.